

This document provides a guide for 3D printing a mount for the OAK-D W camera, designed as an add-on for the Unitree G1 used in the NVIDIA GR00T project.

By providing the accompanying CAD files (.stl) and the brass insert installation guide (.pdf) to a 3D-printing service provider, you can request fabrication of the part.

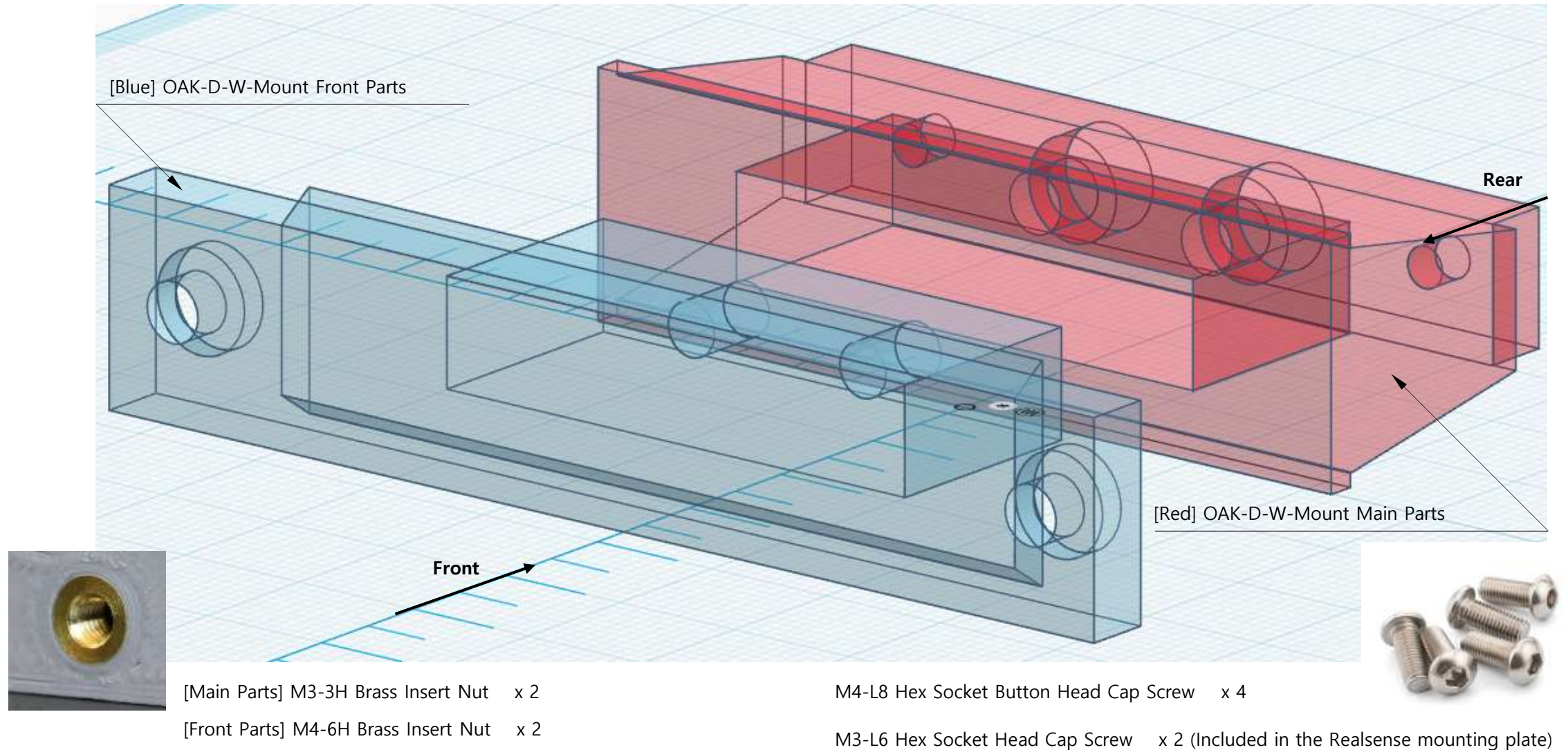
Although the design does not explicitly compensate for printer tolerances, it was successfully manufactured and assembled using a printer with a typical dimensional tolerance of ± 0.2 mm.

The selected manufacturing process was SLA 3D printing with sandblasted surface finishing, using a black ABS-like resin.

Install the brass inserts as shown in the figure on the next page, and prepare the required screws, which must be purchased separately.

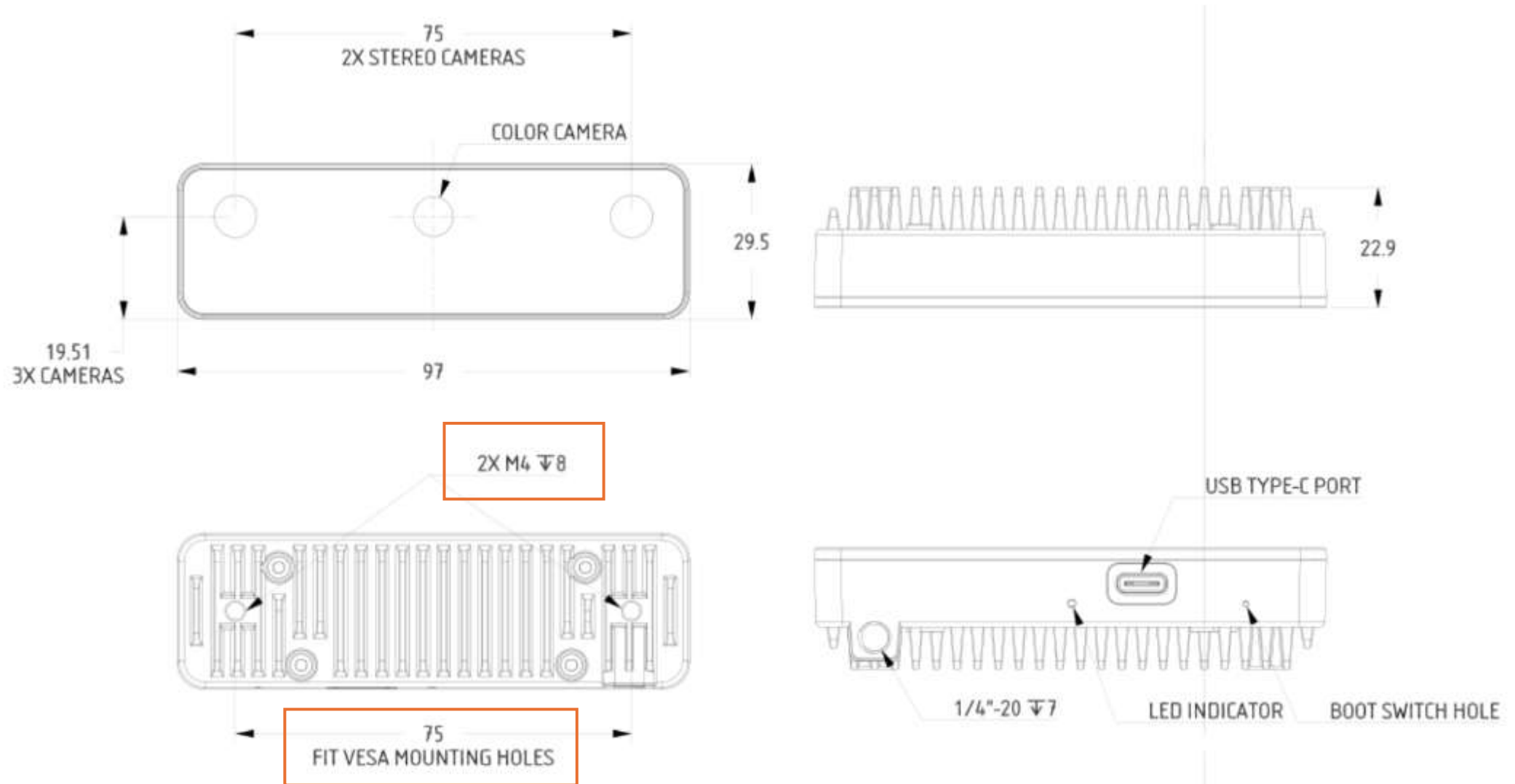
NON-OFFICIAL

OAK-D W Mount



Guide to Attaching an OAK-D W Camera to the Unitree G1 Eye Socket Using a Custom 3D-Printed Mount

OAK-D S2 Camera Mechanical measurements



https://github.com/luxonis/oak-hardware/blob/master/DM9098_OAK-D-S2/Datasheet/OAK-D-S2_Datasheet.pdf

OAK-D-W-Mount Unitree G1 Mount Installation Guide (1)



Secure the Front Mount Parts and the OAK-D W Camera with M4 × L8 Screws



Open the head cover. Disconnect the LED control panel connector, and loosen the LiDAR mounting screws.



Disconnect the RealSense camera cable. Remove the mounting plate screws, and then remove the RealSense camera.



Prepare the OAK-D W camera and the 3D-printed mount.

OAK-D-W-Mount Unitree G1 Mount Installation Guide (2)



Insert the front mount part from the outside of the robot's eye opening and the main mount part from the inside. Align the grooves, then secure the two parts together using M4 × L8 screws.



Once assembled, the mount is held in place by the eye opening. Secure it permanently by fastening it to the camera mounting plate.



Reinstall the LiDAR and reconnect the LED control panel connector. If desired, remove the unused RealSense camera cable by opening the cover on the back of the robot's neck.

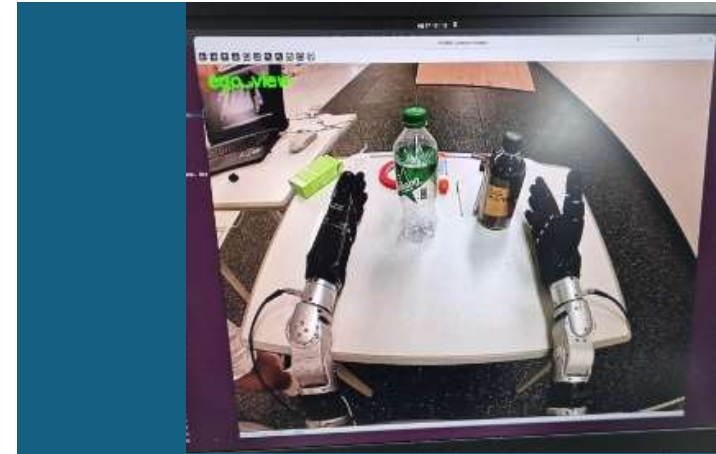


The OAK-D W camera after installation.

RealSense D435i(Left) vs OAK-D W-97(Right)



RealSense D435i Ego Camera View

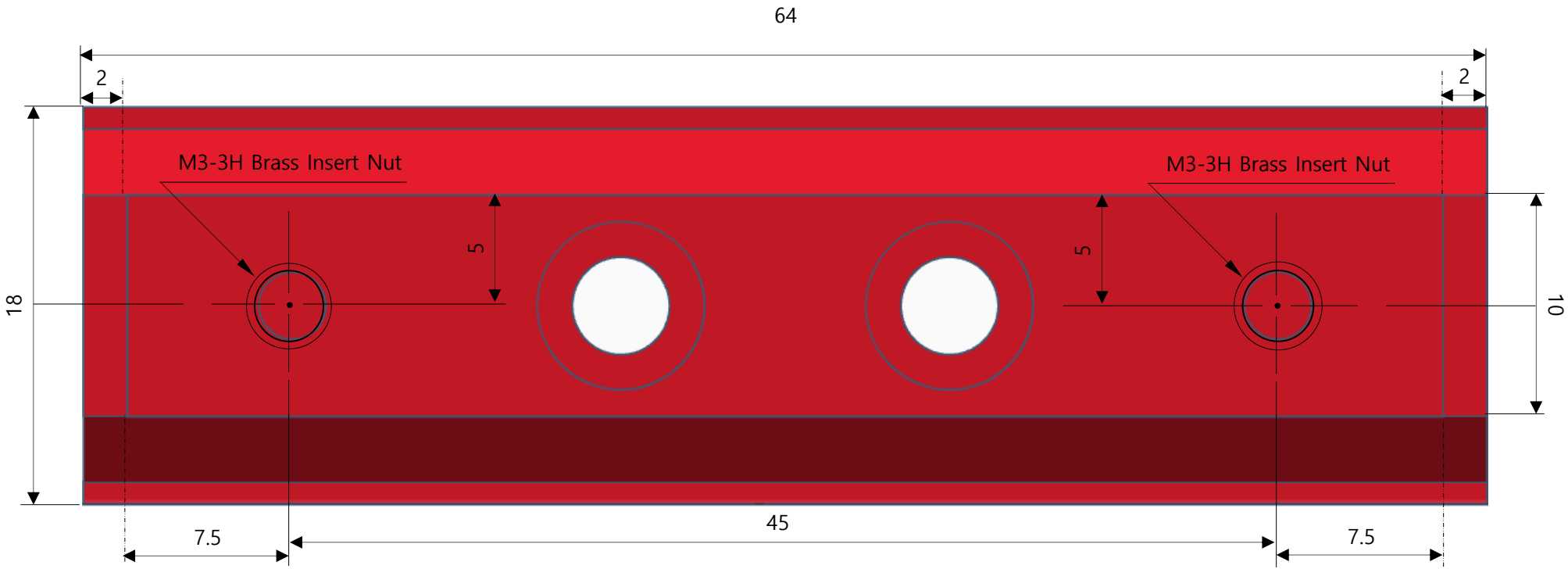


OAK-D W Ego Camera View

Plan View of the Brass Insert Installation Guide Provided to the 3D Printing Service

OAK-D-W-Mount Main Parts Brass Insert Guide

Rear View



OAK-D-W-Mount Front Parts Brass Insert Guide

